

8 Easy Steps to Burn a Perfect CD



- Turn the screensaver off while burning a CD so as to not hamper the flow of data to the CD-Writer.
- Defragment the hard disk before burning a CD.
- If you have a fast Writer but a slow processor and hard disk subsystem (ATA 33), it is safer to burn the CD at a speed lower than the maximum possible.
- Use Track At Once instead of Disk At Once to burn an Audio CD. This will make the Audio CD look more professional because a 2 second gap will be inserted between each burned track and the CD can then be mass produced error-free.
- Always 'finalise' a burned Audio CD to ensure that it plays on your home music system.
- Use branded CDs for burning critical data—cheaper media are prone to damage and can't hold data for long. Plus there is no guarantee that they will burn everything perfectly.
- Shut down all other applications while burning CDs; do not use the computer for any other tasks while a CD is burning.
- Try to simulate a burn process if the Writer has just presented you with a coaster. Most CD-Writing software will allow this.

the Yamaha writer, the tray came out without the CD, which forced us to keep the drive flat on its back to get the media out. This can be harmful for the media as well as the Writer if repeated frequently.

The accessories included in the package add to the overall value of the product. If the package includes the connecting cables, a variety of blank media and a good mix of software, it saves you the expense and bother of buying these separately.

The best combination of accessories was provided by the Samsung SW204—the package included the audio and IDE cables, one blank CD-R disk and one CD-RW disk, Nero 5.5.7.8 and InCD 3.0, and a manual. The ASUS CRW-4816A also comes with the audio and IDE cables, a blank CD-R and CD-RW disk, Nero 5.5.8.0 and a manual. Sony's CRX-195A1 was meagre on accessories—it included only Nero 5.5.8.3 and InCD 3.12.0.

Performance

The proof of a CD-Writer is in its performance. Hence, we tested the speed and reliability with which the CD-Writers could burn CDs, read data off CDs and rewrite data on CD-RWs.

Write tests (CD-R): Though a CD-Writer can double up as a CD-ROM drive, the main function obviously is to burn CDs. Hence we tested this through various tests and benchmarks.

ISO Write Speed: If you need a CD-Writer to back up large chunks of data, such as 200 MB TIFF files, texture files or animation files, then its performance in burning sequential data is what you should look at. To test the sequential data-burning capacity, we used a 700 MB ISO file. This was burnt onto a CD using Nero 5.5.9.14, at the highest speed supported by the CD-Writer. We were expecting the 48x writers to bag this test—the medium after all was a perfect match for their class and the data was of a sequential nature—but nothing of this sort happened. The slower 40x Sony CRX-195A1 surpassed all others here, taking just 152.56 seconds to burn the ISO image. In fact, the runner-up in test was also a 40x Writer, the LITEON LTR-40125 S. It took just 2 seconds more to burn this data, which in practical terms does-

n't make it slower. Krypton BCE-3212M (32x) and Krypton BCE-4012M (40x) burned the ISO image at a really slow 261.44 and 255.61 seconds, respectively.

Assorted Write Speed: To test performance in writing non-sequential data, we burnt a folder containing files ranging from 1 KB to 100 MB onto a blank CD, at the highest speed supported. Here, as expected, the fastest times were logged by the 48x drives. The CD-Writer from Iomega was the fastest, taking just 181.19 seconds to burn the 700 MB file. Close on its heels was ASUS CRW-4816A, taking an insignificant 3 seconds more.

The slowest here was again the Krypton BCE-3212M, taking 343.76 seconds to finish the test—this makes it almost twice as slow as the fastest drive. Iomega is the best buy for those who take a lot of backups on CD-Rs and require a faster drive which can do the task in minimum time.

Audio Write Speed: We tested the speed with which the CD-Writers could write audio on a CD by burning an 80-minute Audio CD from WAV files. Here again the Iomega logged the fastest time by burning the Audio CD in 181.57 seconds—this time is practically the same as the 181.19 seconds it took to burn the assorted (non-sequential) data, which indicates that this drive will give consistent results for different types of data.



Sony's CRX-195A1 took just 3 seconds more to finish writing the Audio CD. The Krypton writers performed poorly in this test as well. The 32x Krypton drive took 340.16 seconds, while its faster sibling, the 40x writer, took 464.96 seconds.

Raw Write Speed: For the Raw write speed test, we attached an extra 48x CD-ROM as the primary master. Using CloneCD, we then created an exact replica of the Digit Test CD. We expected the Writers to take longer here, since the data was being read from a CD-ROM drive, but the Writers did just fine and didn't log any extra time here. The Iomega Writer once again logged an impressive time of 182.89 seconds to finish cloning the CD. The ASUS CRW-4816A was once again a close second, taking just 185.05 seconds. The slowest drive here was again the Krypton BCE-4012M; it took 336.57 seconds.

Rewrite tests (CD-RW): We also evaluated the drives for their rewriting capabilities via the RW writing and erasing tests. These scores are especially relevant today as the RW media is slowly replacing the good old floppy for everything from booting the system to just transferring data from one machine to another.

RW-Assorted Write Speed: We burned the folder containing files ranging from 1 KB to 100 MB on a CD-RW. We tested the Writers using an ASUS 16x media. Although two of the drives, the Iomega and Yamaha, were capable of re-writing data at 24x, the unavailability of appropriate media forced us to go with the a 16x RW CD. The ASUS writer did wonderfully in this test; it finally overtook its closest rival, the Iomega drive, by logging 322.95 seconds, beating it by 7 seconds. The Benq 4012P finished last, taking 530.86 seconds to complete the task at a speed of 12x—

